

Figure 4 | Clones outside the germinal centre

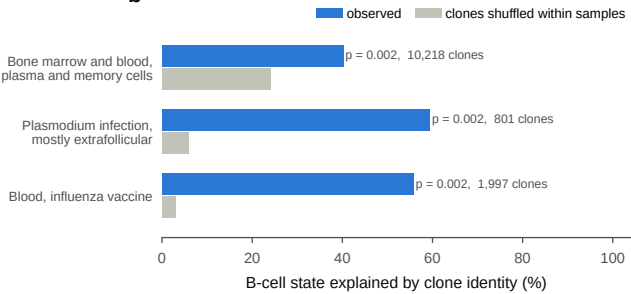
Long-lived plasma cells in human bone marrow, and the first two weeks of a Plasmodium infection, where most responding cells never enter a germinal centre.

a

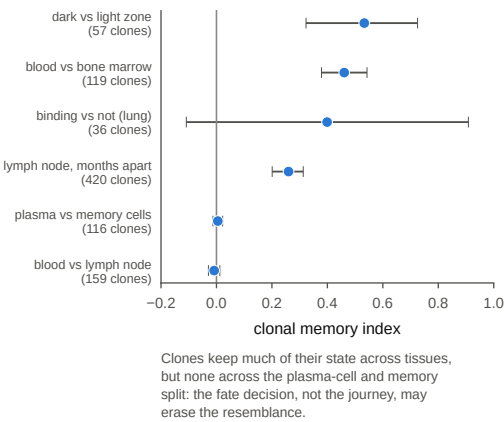
Outside the germinal centre: what a clone is when nothing is selecting it



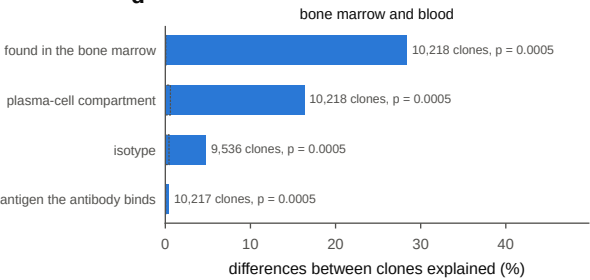
b



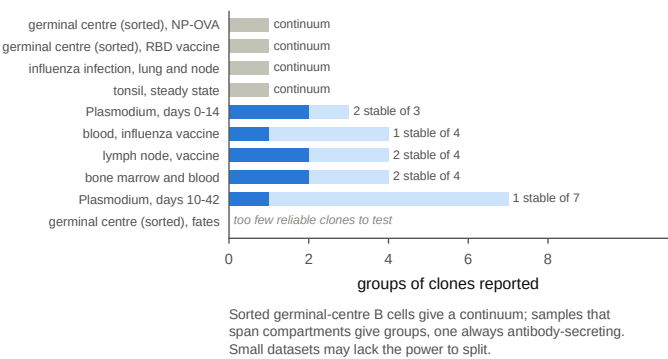
c



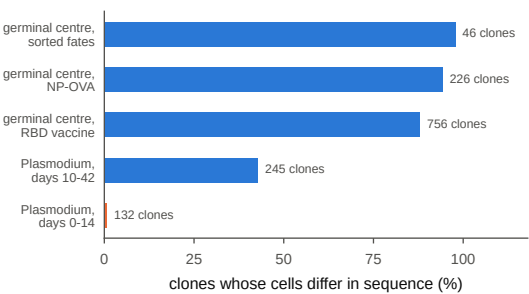
d



e

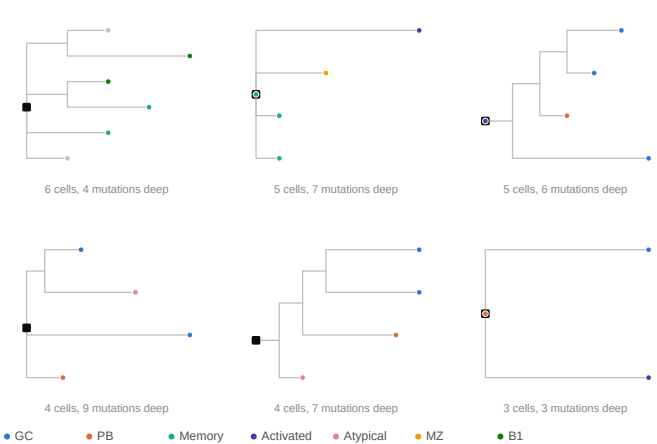


f



g

Lineage trees of barely expanded clones, days 10-42



Each tree is one clone, reconstructed from its heavy-chain V regions with the unmutated ancestor as the root (square); each dot is a cell, coloured by what it was doing. Clones of three to eight cells already hold cells of different fates on different branches.